St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal (CPR) | IMCQ 1

CPR IMCQ 1

CPR iMCQ 1
Physiology Questions

1. A 55-year-old man comes to the physician because of a 4-week history of exertional dyspnea and progressive ankle swelling. He has a history of poorly controlled hypertension and type 2 diabetes. His blood pressure is 150/90 mm Hg, and pulse is 80/min. Physical examination shows jugular venous distention and mild pitting edema of the lower extremities. Right heart cardiac catheterization shows a right atrial pressure of 8 mm Hg and cardiac output of 5 L/min. Which of the following is most likely a best estimate of this patient's total peripheral resistance in mm Hg/L/min?

A. ~ 18

B. ~20

C. ~ 24

D. ~ 26

E. ~ 28

1. A 55-year-old man comes to the physician because of a 4-week history of exertional dyspnea and progressive ankle swelling. He has a history of poorly controlled hypertension and type 2 diabetes. His blood pressure is 150/90 mm Hg, and pulse is 80/min. Physical examination shows jugular venous distention and mild pitting edema of the lower extremities. Right heart cardiac catheterization shows a right atrial pressure of 8 mm Hg and cardiac output of 5 L/min. Which of the following is most likely a best estimate of this patient's total peripheral resistance in mm Hg/L/min?

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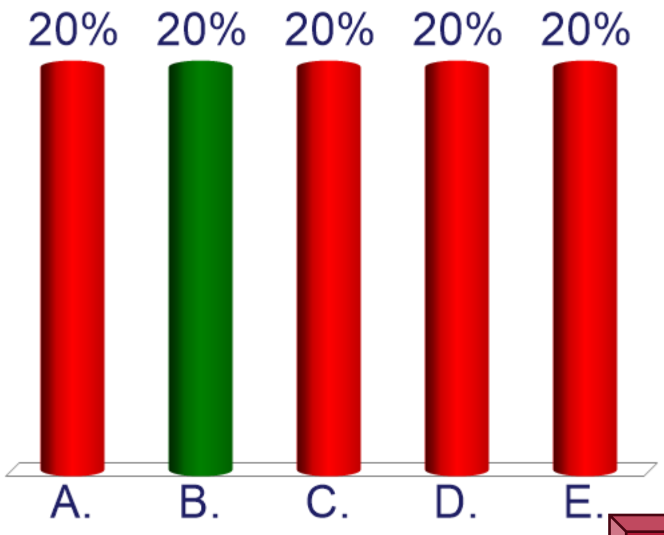
SOM.MK.II.BPM1.3.CPR.2. PHYS.0085: Define cardiac output (CO)and stroke volume (SV). Explain how CO and SV are calculated and be able to list normal value ranges.



2. A 70-year-old man comes to the physician because of a 10-month history of shortness of breath and chest discomfort while climbing stairs. He wakes up at night, gasping for air, and needs multiple pillows to sleep comfortably. He has coronary artery disease and a previous myocardial infarction. Physical examination shows bilateral lower extremity pitting edema, an S3 gallop on auscultation, and elevated jugular venous pressure. An echocardiogram shows a dilated left ventricle with reduced ejection fraction. Hemodynamic tracings of the left ventricular pressure are recorded. Following the timing of this abnormal heart sound, which event is most likely to occur next?

A. Beginning of isovolumetric ventricular contraction
B. Beginning of isovolumetric ventricular relaxation
C. End of isovolumetric ventricular contraction
D. End of isovolumetric ventricular relaxation
E. During rapid filling phase
F. During atrial systole

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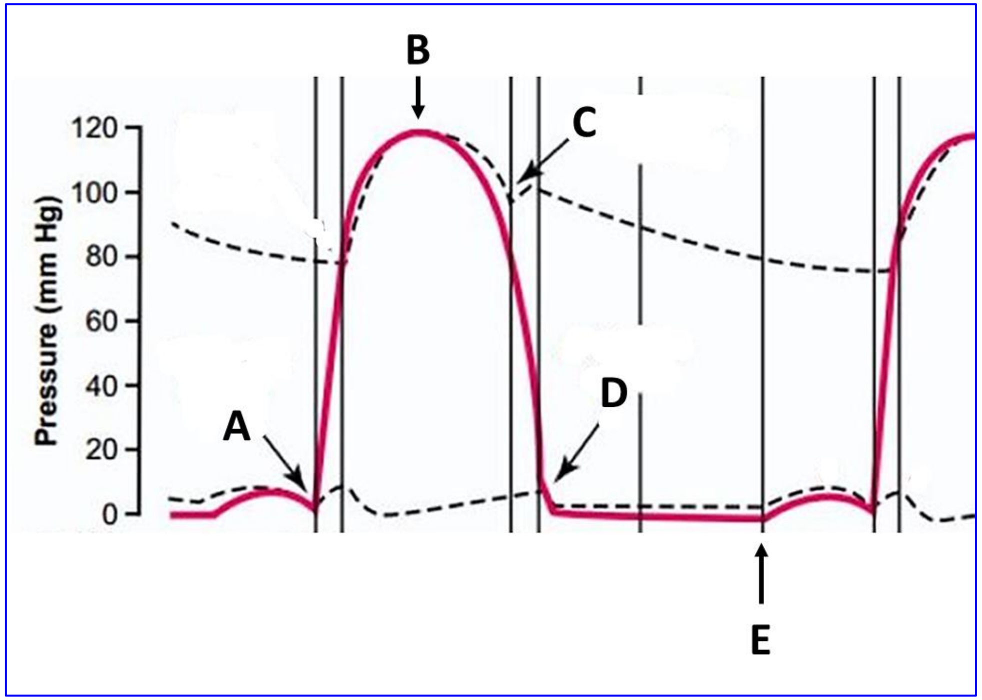
C. End of isovolumetric ventricular contraction
D. End of isovolumetric ventricular relaxation
✓E. During rapid filling phase
F. During atrial systole

SOM.MK.II.BPM1.3.CPR.2.PHYS.0090. Illustrate and explain, in a correct temporal relationship, the phases of the cardiac cycle (atrial systole, isovolumetric contraction, rapid ventricular ejection, reduced ventricular ejection, isovolumetric relaxation, rapid ventricular filling, reduced ventricular filling/diastasis) and the relationship of each phase with the left ventricular pressure curve, the left ventricular volume curve, the left atrial pressure curve, the aortic pressure curve, and the heart sounds (S1, S2, S3, S4).

3. A 41-year-old woman comes to the physician for a follow-up examination. She has a 5-year history of chronic obstructive pulmonary disease. She has smoked one-pack of cigarettes daily for 20 years. Her pulse is 86/min, respirations are 20/min, and blood pressure is 128/84 mm Hg. Physical examination shows a hyperresonant chest and a loud second heart sound in the pulmonary area. Echocardiography shows right ventricular hypertrophy but a normal left ventricle. A left-sided pressure recording is shown in the graph. Which of the following intervals between the two points in the graph best represents an unchanging volume of the left ventricle in this patient?

A. A to B
B. B to C
C. C to D
D. D to E
E. E to A



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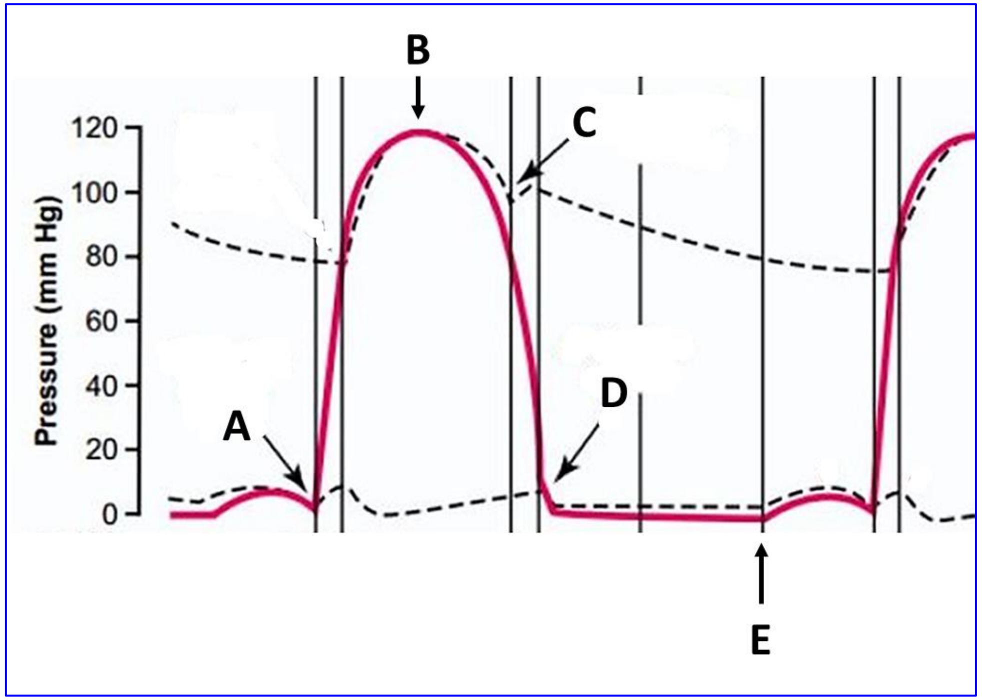
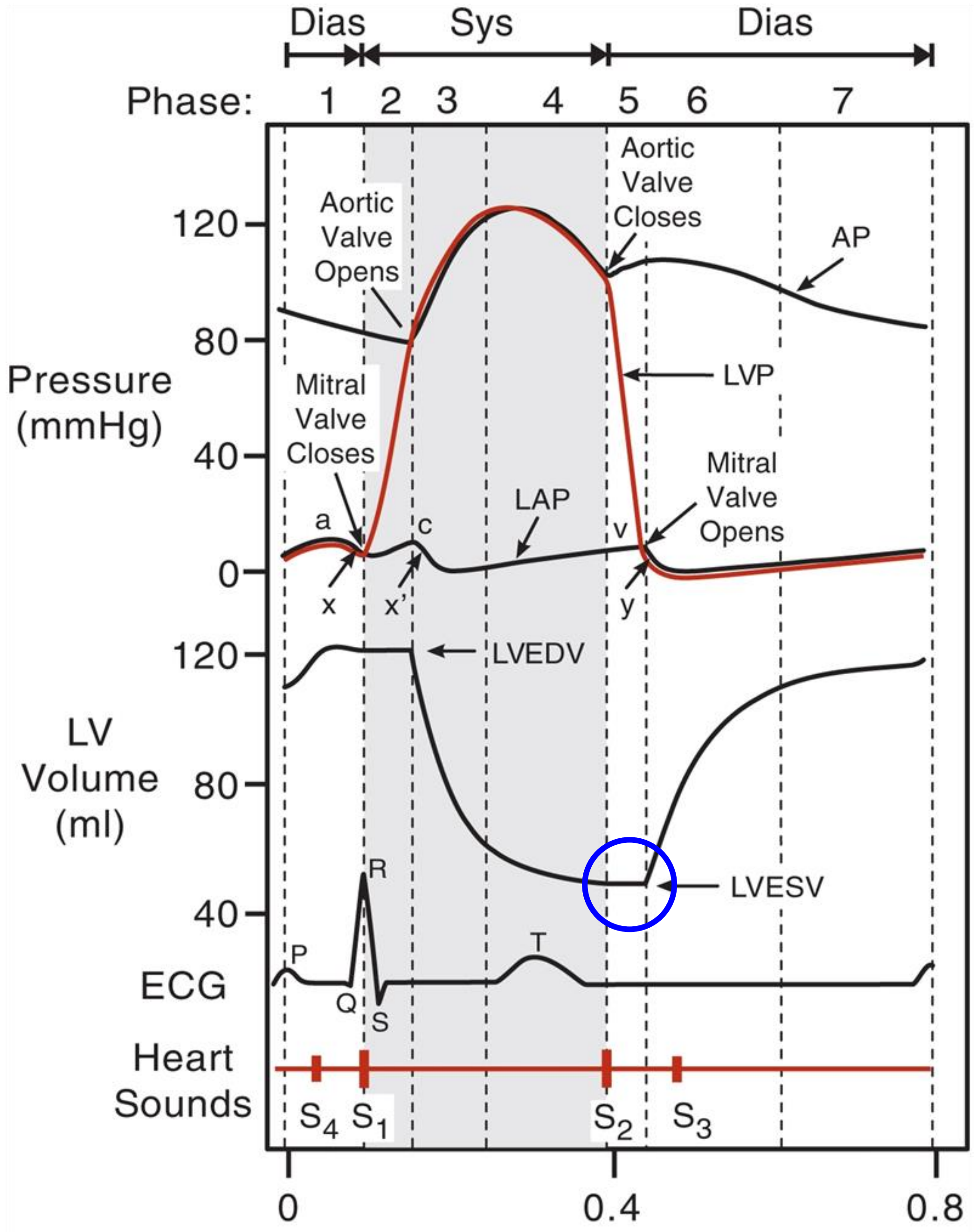
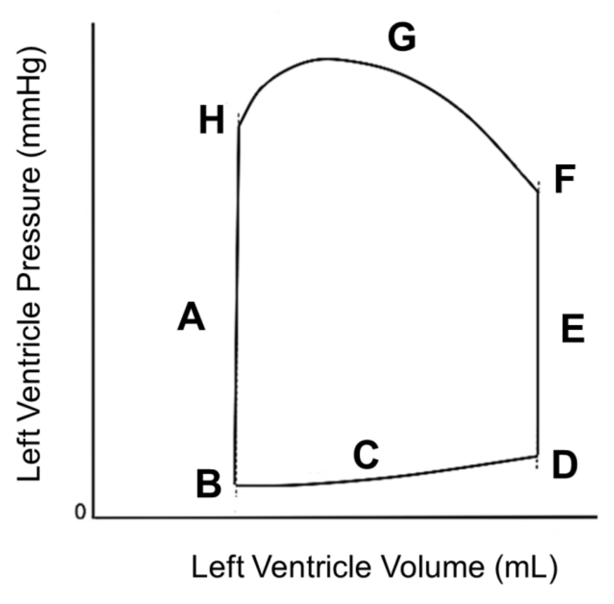


SOM.MK.II.BPM1.3.CPR.2.PHYS.0090. Illustrate and explain, in a correct temporal relationship, the phases of the cardiac cycle (atrial systole, isovolumetric contraction, rapid ventricular ejection, reduced ventricular ejection, isovolumetric relaxation, rapid ventricular filling, reduced ventricular filling/diastasis) and the relationship of each phase with the left ventricular pressure curve, the left ventricular volume curve, the left atrial pressure curve, the aortic pressure curve, and the heart sounds (S1, S2, S3, S4).


Dias
Sys
Dias
Phase:
1
2
3
4
5
6
7
Aortic Valve Opens
Aortic Valve Closes
AP
LVP
Pressure (mmHg)
120
80
40
0
Mitral Valve Closes
Mitral Valve Opens
a
c
v
x
x'
y
LAP
LVEDV
LV Volume (ml)
120
80
40
LVESV
R
P
T
Q
S
ECG
Heart Sounds
S_4
S_1
S_2
S_3
0
0.4
0.8

4. A 63-year-old woman comes to the physician because of a 12-week history of increasing exertional dyspnea and fatigue. She has history of ischemic heart disease and underwent mitral valve repair 5 years ago. Her pulse is 88/min, respirations are 20/min, and blood pressure is 130/82 mm Hg. Cardiac auscultation shows a holosystolic murmur best heard at the apex and radiating to the axilla. Echocardiography shows moderate mitral regurgitation and mild left atrial enlargement. The pressure-volume relationship of this patient's left ventricle during the cardiac cycle is shown in the graph below. Which of the following labeled points in the graph shown most likely corresponds to the time of onset of the murmur in this patient?

A. A
B. B
C. C
D. D
E. H



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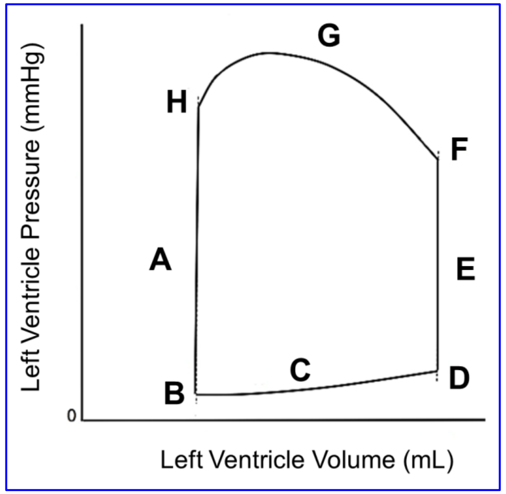

SOM.MK.II.BPM1.3.CPR.2.PHYS.0091. Explain the ventricular volume-pressure relationship during the different phases of the cardiac cycle in the pressure-volume loop, and the points in the graph that correspond to the the mitral valve closure (S1), the mitral valve opening, aortic valve closure (S2), the aortic valve opening, S3 sound, S4 sound, end diastolic volume, and end systolic volume.

HCB Questions

A 78-year-old woman comes to the physician because of a 3-month history of shortness of breath, worse with exertion, increasing fatigue, and mild lower extremity swelling by evening. She has a history of hypertension and chronic kidney disease. Her blood pressure is 130/70 mmHg, and pulse is 62/min. Physical examination shows a harsh, crescendo-decrescendo systolic ejection murmur heard best at the right upper sternal border, crackles at the bases of both lungs, and 1+ pitting edema of the ankles. Imaging studies show a stenosis of the aortic heart valve. Examination of a biopsy specimen of the affected valve shows an abnormal accumulation of a particular extracellular matrix component. Which of the following components of the extracellular matrix is most likely affected resulting in the changes observed in the affected valve?

A. Proteoglycan
B. Glycosaminoglycans
C. Collagen type I
D. Elastic fibers
E. Collagen type III

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D. Elastic fibers
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SOM.MK.I.BPM1.3.CPR.1.HCB.1110 Describe the histological structure of the heart valves.

A 33-year-old woman comes to the physician because of a 1-day history of shortness of breath and a 2-week history of pain and swelling of the right leg and foot. Her blood pressure is 139/79 mm Hg, pulse is 92/min, respirations are 23/min, and pulse oximetry shows an oxygen saturation of 94% at room air. Physical examination of the right lower extremity shows erythema, warmth, and tenderness of the calf, and mild edema of the foot. A vascular ultrasound show a thrombus in the right posterior tibial vein. Laboratory studies show elevated thrombomodulin levels. Which of the following mechanisms associated with thrombomodulin is mostly to prevent clot formation?

A. Inhibition of platelet attachment and aggregation
B. Promoting the secretion of von Willebrand factor
C. Increasing the production of free radicals
D. Increasing the uptake of LDL-cholesterol from the blood
E. Sustained contraction of the vascular smooth muscle within the tunica media

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SOM.MK.I.BPM1.3.CPR.1.HCB.1120 List 3 substances released by endothelial cells that prevent thrombus formation.
SOM.MK.I.BPM1.3.CPR.1.HCB.1118 Discuss the functions of vascular endothelial cells.

A 16-year-old boy is brought to the emergency department because of a 3-hour history of a high temperature and a generalized rash. His temperature is 105°F (40°C). Physical examination shows a lethargic male with a non-blanching rash, neck stiffness, and an inability to tolerate light. He is diagnosed with bacterial meningitis. Despite appropriate medical treatment, he dies. Autopsy shows disruption in the basal lamina of the vessels that form the blood-brain barrier. Which of the following cell types most likely normally functions to support the basal lamina of these vessels and demonstrates contractile properties?

A. Endothelial
B. Vascular smooth muscle
C. Fibroblasts
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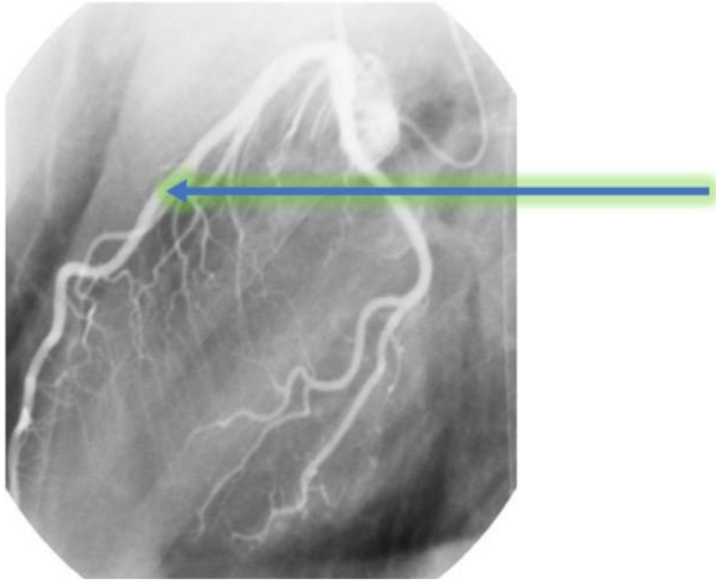
SOM.MK.I.BPM1.3.CPR.1.HCB.1136 Describe the microscopic structure of the three types of capillaries.

A 73-year-old woman is brought to the emergency department because of a 3-hour history of severe left-sided chest pain with profuse sweating and nausea. She has uncontrolled type II diabetes mellitus. Her blood pressure is 80/50 mmHg, respirations are 25/min, pulse is 40/min. She appears in obvious painful and other distress. ECG changes are consistent with a myocardial infarction involving the part of the heart irrigated by the vessel shown in the attached image. Which of the following is most likely a characteristic feature of this vessel?

A. Well-defined scalloped appearance to the tunica intima
B. Continuous basal lamina
C. Tunica media rich in elastic fibers
D. Well-developed tunica adventitia with smooth muscle fibers
E. Wall contains a core of loose connective tissue



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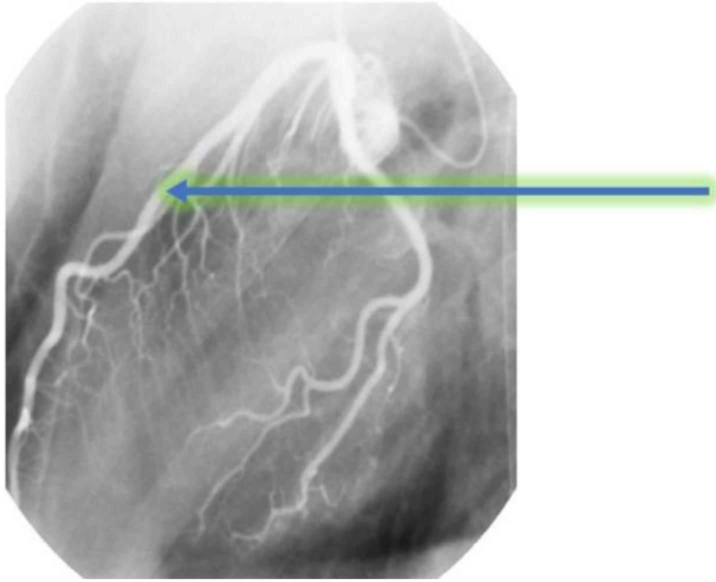

SOM.MK.I.BPM1.3.CPR.1.HCB.1131 Describe the microscopic structure of medium (muscular) arteries.

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Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal (CPR) | IMCQ 1

CPR IMCQ 1
Anatomy

1. A 75-year-old man comes to his physician because of a 4-month history of dizziness, fatigue, and shortness of breath. He has a history of hypercholesterolemia. He has smoked one pack of cigarettes daily. He is 172 cm (5 ft 8 in) tall, weighs 80 kg (176 lb); BMI is 27.0. His temperature is 36.9°C (98.4°F), pulse is 88/min, respirations are 18/min, and blood pressure is 150/85 mm Hg. Physical examination shows a slow-rising, weak pulse along with a prominent systolic murmur best heard at the second right intercostal space that radiates to the carotid arteries. There is no jugular venous distension. The lungs are clear. Which of the following is the most likely diagnosis?

A. Mitral valve prolapse

B. Tricuspid stenosis

C. Pulmonary stenosis

D. Aortic stenosis

E. Mitral stenosis

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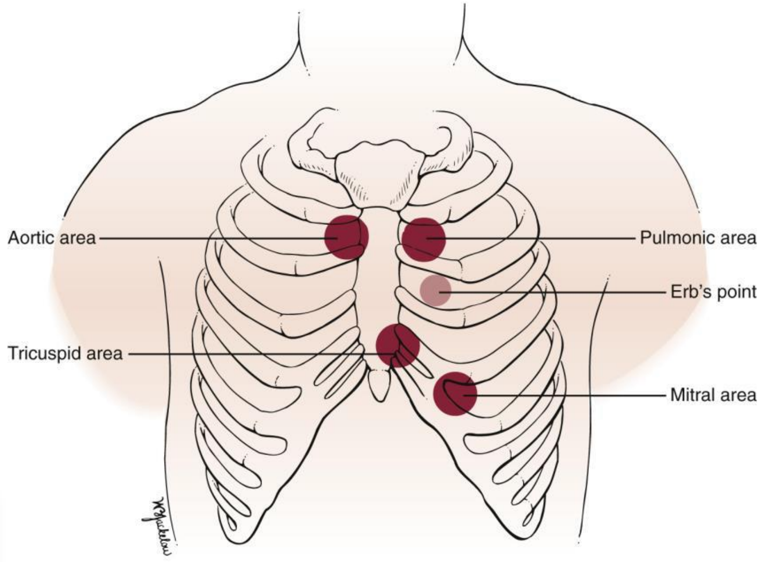
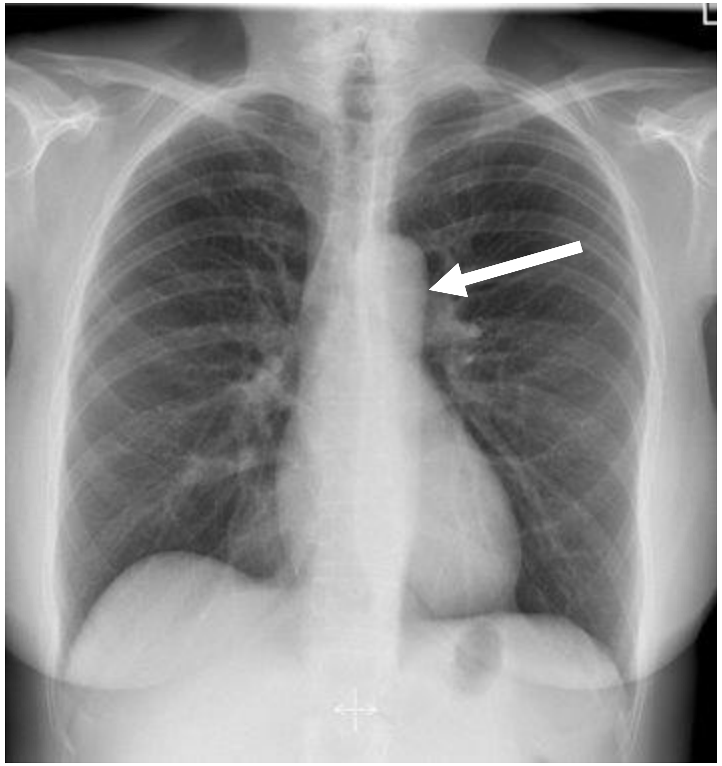


SOM.MK.II.BPM1.3.CPR.2.ANAT.1391 Identify the auscultation sites for heart valve sounds and explain why these auscultation points differ from the anatomical location of the valves.

2. A 42-year-old man comes to the emergency department because of a 2-month history of hoarseness of voice. His temperature is 36.9°C (98.4°F), pulse is 88/min, respirations are 18/min, and blood pressure is 150/85 mm Hg. Physical examination shows clear breath sounds bilaterally. A CT scan of her chest shows a mass located in the mediastinum as indicated by arrow in this normal chest x-ray. Which of the following nerves is most likely being compressed?



A. Right recurrent laryngeal nerve
B. Left recurrent laryngeal nerve
C. Right phrenic
D. Left phrenic
E. Left vagus
F. Right vagus
G. Greater thoracic splanchnic

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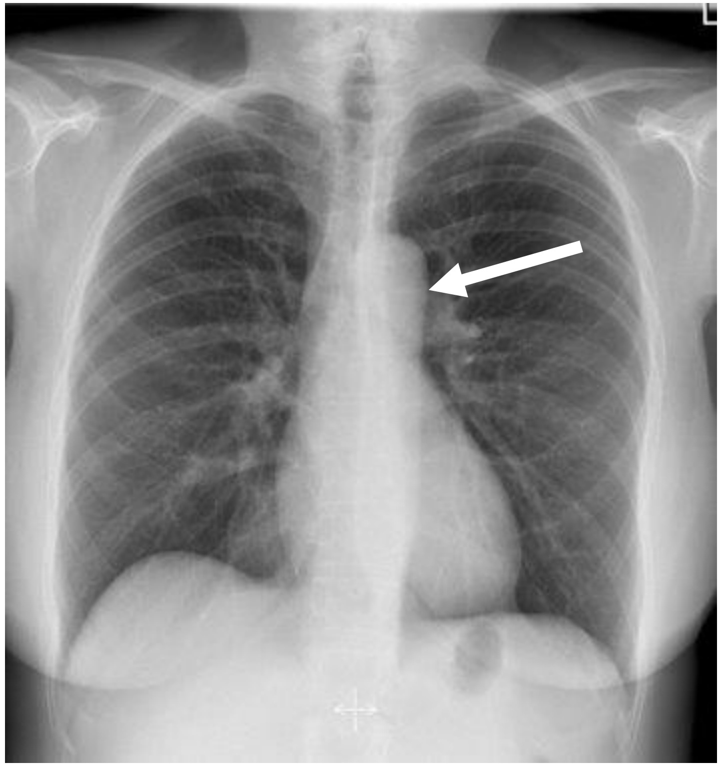

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F. Right vagus
G. Greater thoracic splanchnic

SOM.MK.II.BPM1.3.CPR.2.ANAT.1371	Describe the course of the right and left recurrent laryngeal nerves in the thorax.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1351	Identify the major anatomical features of the mediastinum, heart and great vessels in radiographic images, CT and MRI.

3. A 78-year-old woman comes to the physician because of a 1-week history of swelling of the left upper limb. She reports a feeling of heaviness in her left upper limb. Her medical history includes hypertension and a previously identified aneurysm of the aortic arch. Her pulse is 88/min, respirations are 16/min, and blood pressure is 140/85 mmHg. Physical examination shows visible edema in the left upper arm, with normal skin color and temperature. Pulses are palpable and strong bilaterally. Further assessment of her chest shows a faint bruit over the upper left thorax. Bedside ultrasonography and subsequent CT imaging confirm an aortic arch aneurysm compressing a large vein located anterior and superior to the aneurysm. Which of the following veins is most likely being compressed?

A. Azygos
B. Internal thoracic
C. Left brachiocephalic
D. Left superior intercostal
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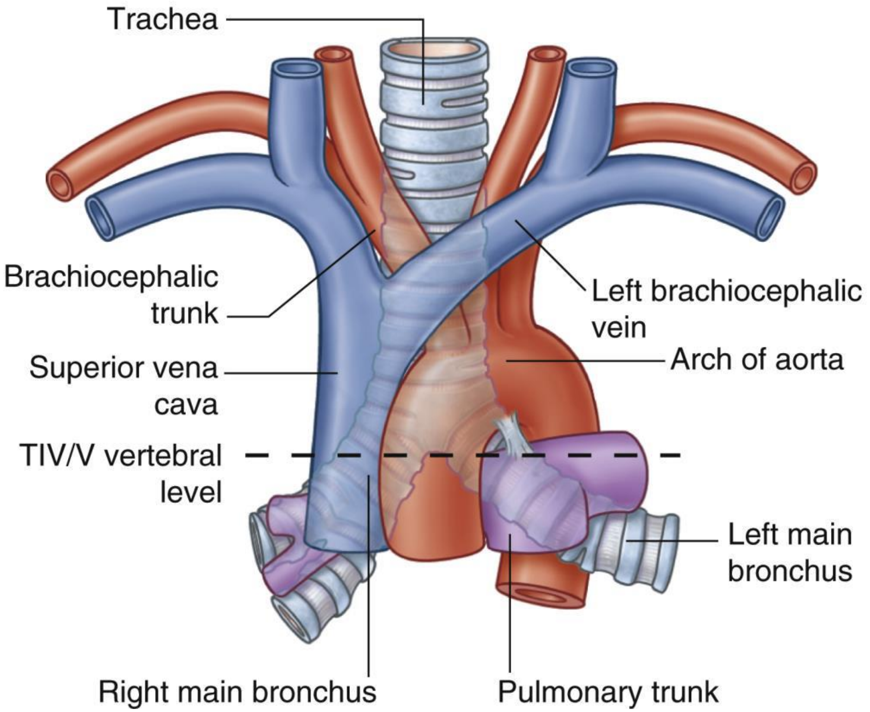

SOM.MK.II.BPM1.3.CPR.2.ANA T.1364 List the structures found in the superior, anterior, middle and posterior mediastinum and their relationships to one another.

4. A 56-year-old man is brought to the emergency department with a 2-hour history of sudden onset of severe chest pain. His pulse is 86/ min, respirations are 16/min, his blood pressure is 90/60 mm Hg. He appears to be in severe distress, with distended jugular veins. Cardiac auscultation shows no murmurs or extra heart sounds. The lungs are clear to auscultation bilaterally. An ECG shows ST-segment elevation consistent with a myocardial infarction. Bedside echocardiography shows reduced contractility of the right ventricle, while the left ventricle exhibits normal function. Coronary angiography shows a thrombotic occlusion affecting one of the major coronary arteries, and absence of significant atherosclerotic lesions. Which of the following best explains the findings in this patient?

	Occluded artery	Coronary artery dominance
A.	Right coronary	Right dominant
B.	Right coronary	Left dominant
C.	Anterior Interventricular	Right dominant
D.	Anterior Interventricular	Left dominant
E.	Left circumflex	Right dominant
F.	Left circumflex	Left dominant

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E.	Left circumflex	Right dominant
F.	Left circumflex	Left dominant

SOM.MK.II.BPM1.3.CPR.2.ANAT.1393 Explain which part of the myocardium will be affected if any of the coronary arteries is occluded.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1394 Describe the concept of left and right dominance as it relates to the blood supply of the heart.

5. A 56-year-old man is brought to the emergency department with chest pain and shortness of breath that began suddenly while he was at rest. He describes the pain as severe with a sensation of pressure, radiating to his left inner arm. He reports occasional mild chest discomfort over the past few months, which he had attributed to stress. He has a history of hypertension and high cholesterol but has been otherwise healthy. The patient is 178 cm (5 ft 10 in) tall and weighs 85 kg (187 lb), BMI is 26.8. His temperature is 37.0°C (98.6°F), pulse is 112/min, respirations are 22/min, and blood pressure is 154/88 mm Hg. Physical examination shows diaphoresis and pallor. Cardiac auscultation shows an S4 gallop, but no murmurs are heard. Laboratory studies show elevated cardiac biomarkers. An electrocardiogram shows ST-elevation in leads V1, V2, V3, V4. Cardiac catheterization shows occlusion of an artery that supplies 2/3 of the interventricular septum. Which of the following coronary arteries is most likely occluded in this patient?

A. Anterior interventricular artery (Left anterior descending "LAD")
B. Circumflex branch of the left coronary artery (LCX)
C. Right coronary artery (RCA)
D. Posterior (inferior) interventricular artery (Posterior descending artery "PDA")
E. Left main coronary artery

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SOM.MK.II.BPM1.3.CPR.2.ANAT.1394 Describe the concept of left and right dominance as it relates to the blood supply of the heart.

6. **A 70-year-old man is brought to the emergency department because of a 2-day history of high fever, chills, and chest pain. Physical examination shows a holosystolic murmur that radiates toward the axilla. Two sets of blood cultures grow Staphylococcus aureus. A transesophageal echo is performed, and the ultrasound probe is placed in the mid-esophagus facing posteriorly. Which of the following will be best visualized in this position?**

A. Left atrium
B. Left ventricle
C. Right atrium
D. Right ventricle
E. Descending aorta
F. Pulmonary artery
G. Pulmonary veins
H. Superior vena cava

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G. Pulmonary veins
H. Superior vena cava

SOM.MK.II.BPM1.3.CPR.2.ANAT.1365 Describe the relationships of the esophagus, trachea and thoracic duct in the superior mediastinum.

7. A 65-year-old man is brought to the emergency department by his wife because of intense and unremitting chest pain for the past 1 hour. He describes the pain as retrosternal, radiating to the medial aspect of his left arm. There is no family history of heart disease. His temperature is 37°C (98.6°F), pulse is 120/min and blood pressure is 130/80 mm Hg. ECG shows a myocardial infarction affecting the anterolateral left ventricular wall. An x-ray of the chest shows no abnormalities. Which of the following nerves is most likely responsible for the arm pain felt by this patient?

A. Vagus
B. Thoracic visceral afferents
C. Phrenic nerve
D. Intercostobrachial
E. Cardiopulmonary

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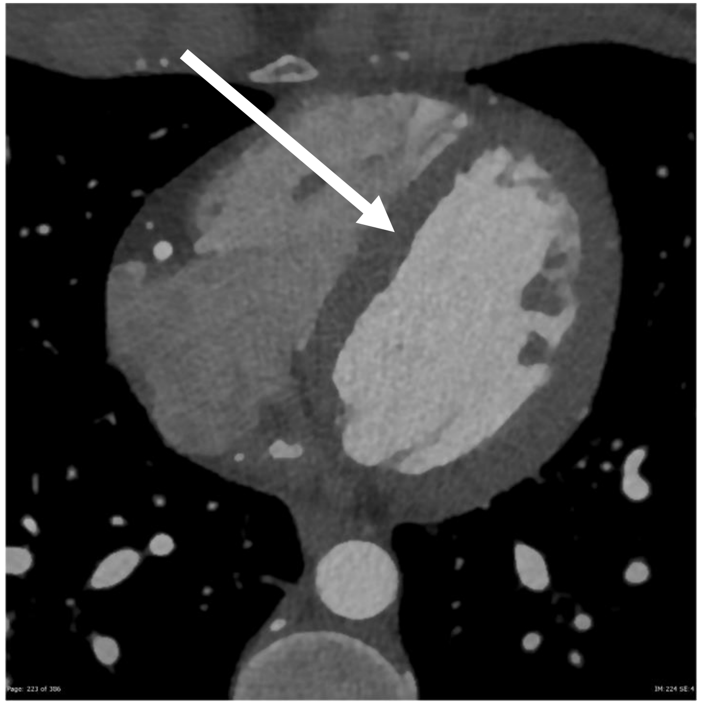
SOM.MK.II.BPM1.3.CPR.2.ANAT.1400 Describe the intercostobrachial nerve and its clinical significance in relation to the upper limb and myocardial infarction.

8. A 65-year-old man is brought to the emergency department by his wife because of a 1-hour history of intense and unremitting chest pain. He describes the pain as retrosternal, radiating to the left side of his neck and shoulder and down to the medial aspect of his left arm. There is no family history of heart disease. His temperature is 37°C(98.6°F), pulse is 120/min and blood pressure is 130/80 mm Hg. ECG shows a myocardial infarction. Coronary angiography shows nearly total blockage of the artery lying on the anterior aspect of the structure as indicated by the arrow. When this artery is exposed for a coronary bypass graft procedure, which of the following structures is at greatest risk for injury in this patient?



A. Middle cardiac vein
B. Small cardiac vein
C. Posterior interventricular artery
D. Great cardiac vein
E. Anterior cardiac vein
F. SA nodal artery
G. Circumflex artery

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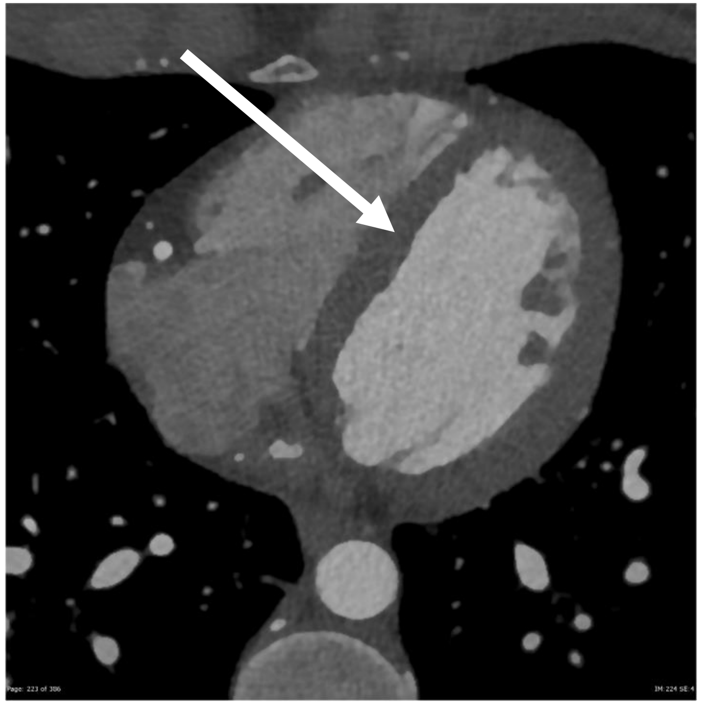

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F. SA nodal artery
G. Circumflex artery

SOM.MK.II.BPM1.3.CPR.2.ANAT.1392 Describe the arterial supply and venous drainage of the heart.